CAST

Product Bible

Powered by Gravity Technology

Sitemap · Personas · User Workflows · Edge Cases · Feature Register

The complete product specification for Cast V1 → V3

Gravity Technology · Confidential · 2025

C O N T E N T S

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CHAPTER 01

Master Sitemap

Cast is structured as four sovereign zones. Each zone has a distinct purpose, audience, and technical profile. No zone bleeds into another — the separation is architectural, not cosmetic.

Total zones	**4 — Trap, Orbit, Syndicate, Architect**
Total unique routes	**47 primary routes + dynamic segments**
Auth boundary	**All /app/* and /creator/* require JWT**
Public surface	**Zone 1 only — maximum SEO and viral capture**
Rendering	**Next.js Edge for Zone 1 · SSR for Zone 2 · CSR for Zone 3–4**

Zone 1 · The Trap

Public. Unauthenticated. Fully edge-cached. This is the viral surface area — every WhatsApp link, every SEO campus page, every blurred reveal that forces an install. The Trap exists to convert curiosity into accounts.

> **THE TRAP — Public / Unauthenticated**
> cast.gg / cast.gg/vote / cast.gg/war / cast.gg/deck
> **Goal:** Convert visitors into authenticated users. No other metric matters here.

Zone 1 Routes		
Route + Name	**Purpose + Behaviour**	**Backend Notes**
/ — Genesis Page	Cinematic landing. Live global vote counter. CTA: Enter the Court.	Edge cached · Cloudflare · 0ms TTL drift
/vote/[case_id] — Blurred Reveal	The WhatsApp link. Target's verdict shown blurred. CTA: Download to Unlock. This is the highest-converting page in the product.	AuditService + SharecardService. OpenGraph image pre-generated.
/war/[region_id] — Campus Leaderboard	SEO public page. Top 10 most Casted squads per	LeaderboardService via Redis ZSETs. Static

	university. Updates every 15 min.	HTML with ISR.
/deck/[deck_slug] — Creator Preview	Public deck landing. Shows 3 preview questions. CTA: Play this Cast.	CortexService pre-screens deck before slug is published.
/legal — Terms & Privacy	Full ToS, Privacy Policy, COPPA/GDPR notice, Cameroonian data law compliance.	Static. Legal review required per jurisdiction.
/auth/phone — OTP Entry	Passwordless entry. WhatsApp OTP preferred, SMS fallback.	IdentityService. Rate-limited: 5 attempts / 10 min per device_hash.
/auth/identity — Handle + Avatar	Claim username. Upload or auto-generate avatar. Handle reservation.	IdentityService. Handle locked for 24h if abandoned mid-flow.
/auth/sync — Contact Sync	Mandatory contact graph sync. Encrypted client-side before upload.	Go Worker builds social graph edges async. User enters app immediately.

Zone 2 · The Orbit

The core product. Authenticated users only. A 5-tab mobile-first PWA designed around a single daily ritual — open Cast, see who voted you, vote someone else, share the result. Every pixel is in service of this loop.

Tab 1 · Home — The Habit Engine

Tab 1: Home routes		
Route	**Feature**	**Backend**
/app/home	Daily stack. Pending verdicts badge. Daily Chaos Prompt from Hermes. Check-in streak display.	StreakService + CortexService. Hermes prompt pre-generated nightly at 00:00.
/app/home/daily-prompt	Full-screen daily ritual. Swipe to answer. Squad-wide results revealed after all members vote.	CortexService. Prompt tied to squad calendar event if detected.
/app/home/pending	All unopened verdicts awaiting the user. Sorted by emotional intensity score.	BurnbookService. Intensity score = recency × voter relationship weight.
/app/feed/hotlist	Scoreboard feed. Trending users, global squad wars, most savage titles today.	FeedService. Redis-cached. Refreshes every 5 min.

Tab 2 · Squads — The Party Engine

Tab 2: Squads routes		
Route	**Feature**	**Backend**
/app/squads	List of active squads. Create or join. Unread badge per squad.	SquadsService. Max 10 squads per user in V1.
/app/squads/[id]/lobby	Group chat (text + voice notes). Active decks queued. Online presence dots.	WebSockets via NATS PubSub. Voice stored on R2/S3. Max 30 members.
/app/squads/[id]/leaderboard	Intra-squad rankings. Aura score (positive votes) vs Toxicity score (negative).	Redis ZSET: zset:leaderboard:squad:{id}:aura and :toxic
/app/squads/[id]/burn-book	Collective squad vote history. Every title ever awarded. Sortable by date/type.	BurnbookService materialised view. Snapshot rebuilt on every new vote.
/app/squads/[id]/espionage	Spy on a rival squad's stats. Costs 200 Coins. Shows their top 5 titles and Cast Score.	WalletService deducts + BurnbookService grants 24h read access.
/app/squads/new	Create squad. Name it. Set invite link. Choose public/private.	SquadsService. Invite link = cryptographic token. Expires 7 days if unused.
/app/squads/join/[token]	Join via deep link. Shows squad preview. Requires auth.	SquadsService. Token validated + invalidated on join.

Tab 3 · The Courtroom — The Core Ritual

Tab 3: Courtroom routes		
Route	**Feature**	**Backend**
/app/decks	Deck library. Categories: Chaos, Love & Lies, Campus, Money, Creator, Sponsored.	DeckService. Sponsored decks injected by CampaignService. Creator decks ranked by play rate.
/app/courtroom/[deck_id]	Active voting. Face-grid UI. Tinder-style tap to Cast. Haptic + audio feedback on each vote.	VotingService via NATS Event Bus. Writes async. Returns 200 OK in <10ms.
/app/courtroom/execution	Post-vote cinematic	SharecardService queued

	reveal. Verdict Sealed animation. Share and continue options.	async. Deck completion triggers BurnbookService update.
/app/courtroom/results/[session_id]	Full squad results after all members complete the deck. Aggregate tallies.	VotingService aggregation. Unlocked free — only individual identity is paywalled.

Tab 4 · Arena — Tournaments & Discovery

Tab 4: Arena routes		
Route	**Feature**	**Backend**
/app/leaderboards	Global and regional rankings. Aura and Toxicity separated. Filter by campus/city.	LeaderboardService Redis ZSETs. Global: zset:leaderboard:global:aura
/app/tournaments	Browse upcoming and live tournaments. Entry fee, prize pool, and participant count shown.	TournamentService. Prize pool held in EscrowService until winner confirmed.
/app/tournaments/[id]/live	Live scoreboard during active tournament. Updates every 10s.	WebSocket push from TournamentService.
/app/tournaments/[id]/join	Pay entry fee in Coins. Confirm squad eligibility. Lock-in CTA.	WalletService + EscrowService. Idempotency key required.
/app/tournaments/[id]/results	Final results and prize distribution confirmation.	TournamentService triggers PayoutService on completion.

Tab 5 · Legacy & Wallet — Identity and Economy

Tab 5: Legacy routes		
Route	**Feature**	**Backend**
/app/profile/[user_id]	Public profile. Cast Score. Active titles. Archetype badge. Recent activity.	ProfileService. Cached snapshot. Respects privacy shield.
/app/profile/[id]/burn-book	Full vote history against user. Free tier: count shown, content blurred. Paid: fully unlocked.	BurnbookService. Blur applied client-side with server-validated unlock token.
/app/profile/[id]/trophies	Earned badges: Certified Villain, Streak Lord, Top Casted, etc.	BadgeService. Auto-awarded on threshold events.
/app/profile/[id]/ai-	Hermes archetype card. Full	CortexService. Recalculated

persona	psychological profile. Shareable.	every 7 days or on vote threshold.
/app/wallet	Coin balance. Transaction history. Buy and withdraw CTAs.	WalletService. Double-entry ledger. All reads from Postgres replica.
/app/wallet/buy	Coin packs via MTN MoMo, Orange Money, Paystack.	PaymentService. Idempotency key on every transaction.
/app/wallet/withdraw	Withdraw creator or tournament earnings. KYC gate.	PayoutService. KYCService approval required for first withdrawal >5,000 CFA.
/app/inner-court	Premium upgrade UI. Plan comparison. Monthly / annual / Chairman Pass.	SubscriptionService. Integrates with PaymentService.
/app/settings	Privacy mode, notification toggles, contact re-sync, account deletion.	IdentityService + NotificationService.

Zone 3 · The Syndicate

Web-first portals for power users: creators building decks, campus ambassadors farming referrals, and brands running campaigns. These are not casual users — they are the supply side of the Cast economy.

Zone 3: Syndicate routes		
Route	**Feature**	**Backend**
/creator/dashboard	Analytics hub. Deck play counts, earnings, follower growth, top-performing questions.	CreatorService. Revenue split: 30% creator / 70% platform.
/creator/builder	Deck creation tool. Add questions, set categories, upload cover image. Moderation scan on submit.	CortexService pre-screens via Hermes before publish. Shadowban if flagged.
/creator/analytics/[deck_id]	Per-deck analytics. Play rate, unlock rate, revenue per play, top campuses.	AnalyticsService. Updated hourly.
/creator/earnings	Revenue ledger. Pending vs cleared. Withdrawal trigger.	CreatorService + PayoutService.
/syndicate/dashboard	Campus ambassador hub. Unique referral link. Install tracking. Payout history.	AffiliateService. Conversion event: new user creates squad

		with 5+ members.
/syndicate/metrics	Unique installs, squad creations, active user rate from referrals.	AffiliateService. Event sourced from IdentityService + SquadsService.
/syndicate/payout	1-click MoMo withdrawal for affiliate commissions.	PayoutService. KYC required >10,000 CFA.
/enterprise/dashboard	B2B HR portal. Private squad management. Session scheduling. Sentiment analytics.	EnterpriseService. Fully isolated multi-tenant schema.
/enterprise/internal-squads	Create and manage private squads. Safe Mode enforced. Results aggregated only.	EnterpriseService. No individual vote data ever exposed to managers.
/enterprise/analytics	Sentiment trends across sessions. Department-level aggregates.	AnalyticsService enterprise namespace.
/sponsor/campaigns	Brand campaign upload. Deck injection targeting by campus/region. Budget deposit.	CampaignService. Sponsored decks labelled at UI layer.
/sponsor/analytics/[campaign_id]	Impressions, engagement rate, unlock conversions for sponsored decks.	CampaignService + AnalyticsService.

Zone 4 · The Architect Console

God-mode. Desktop only. You and Hermes see the entire civilisation from here — live revenue, dormant squads, toxicity levels, flagged content, and every financial transaction.

Zone 4: Architect routes		
Route	**Feature**	**Backend**
/architect/command	Live geographic heat map of votes. Real-time DAU. MoMo revenue ticker. Global toxicity index.	All metrics via Redis + ClickHouse analytics DB. WebSocket dashboard.
/architect/cortex	Hermes AI control panel. Aggression/humour sliders. Live provocation feed.	CortexService gRPC admin interface.
/architect/cortex/prompt-engineering	Adjust Hermes' system prompt variables. A/B test provocation	CortexService. Version-controlled prompts.

	templates.	Rollback available.
/architect/cortex/provocation-logs	Full log of every push notification Hermes sent today. Open rate per notification.	NotificationService logs. Hermes gRPC call history.
/architect/cortex/vector-memory	GraphRAG view of squad clustering. What Hermes knows about specific squads.	Qdrant vector DB browser interface.
/architect/treasury	Master ledger. Escrow monitor. Total revenue by day/campus/country.	WalletService + EscrowService. Double-entry read-only view.
/architect/treasury/payout-queue	Approve bulk withdrawals for creators and ambassadors. Batch MoMo trigger.	PayoutService. Manual approval gate for amounts >50,000 CFA.
/architect/treasury/escrow	Active tournament prize pools. Locked funds. Release controls.	EscrowService.
/architect/dungeon	Trust & Safety command. AI-flagged content queue. User reports.	ModerationService.
/architect/dungeon/moderation-queue	Flagged custom decks. Manual approve/reject with reason code.	CortexService flags → ModerationService queue.
/architect/dungeon/kyc-approvals	ID document review for withdrawal-eligible users.	KYCService. Documents stored encrypted, auto-deleted post-approval.
/architect/dungeon/ban-hammer	Shadowban (content invisible but user unaware) or IP/device ban.	ModerationService. Shadowban state stored in Redis.

CHAPTER 02

The 9 Personas

Cast is a multi-sided platform. Its power comes from the tension between nine distinct actor types who each want something different from the system — and whose desires happen to make each other's experience richer. Understanding each persona at depth is not a branding exercise. It is an engineering requirement.

01
The Civilian

The fuel. The data engine. The majority.

DEMOGRAPHICS Age 17–26. University students or recent graduates. Cameroon, Nigeria, Ghana. Smartphone primary internet user. 500MB–2GB daily data budget. Group chats are their social operating system.	**PSYCHOLOGY** Highly attuned to social belonging and public perception. Experiences FOMO as a physical sensation. Defines self-worth through peer validation. Will do more to avoid being excluded than to gain status.
CORE DRIVERS Curiosity about what peers really think. Fear of being the last to know. Desire to belong and be seen within their squad. Boredom during gaps in WhatsApp activity.	**DEEPEST FEARS** Being excluded from a conversation about themselves. Seeing a blurred result and not having the Coins to unlock it. Watching their Cast Score drop without knowing why.
JOURNEY IN CAST Discovers Cast via WhatsApp Status or a direct referral link. Creates an account in under 2 minutes. Syncs contacts. Joins or creates a squad. Votes in their first deck. Receives their first blurred burn book entry. Either pays to unlock or recruits a friend for Coins.	**VALUE THEY BRING** Volume. The Civilian is the raw material that makes every other persona's experience valuable. Without Civilians, creators have no audience, brands have no reach, and Whales have no status to flex.
HOW CAST MONETISES THEM Blurred Burn Book unlock (500 CFA). Coin purchase for espionage or tournament entry. Conversion to Inner Court when sufficiently paranoid.	**RETENTION MECHANISM** Pending verdicts badge. Daily Chaos Prompt. Hermes dormant-squad wakeup notifications. Streak mechanics. Social pressure from squad activity feed.

SUCCESS METRICS

DAU, D1/D7/D30 retention, average sessions per day, first paid conversion rate, referral k-factor.

02
The Inner Court Member

The whale. The status buyer. The one who cannot afford not to know.

DEMOGRAPHICS Age 21–35. Working professional or final-year student. Disposable income. High social visibility on campus or in their industry. Uses Cast as a mirror for their social reputation.	**PSYCHOLOGY** Status-driven to an extreme degree. Paying for Inner Court is not about features — it is about being the person in the squad who always knows. The information asymmetry is the product. They will renew just to avoid losing the immunity shield.
CORE DRIVERS Total information: knowing every verdict, every voter pattern, every title. Status signals: the Elite badge tells the squad they operate at a different level. Control: the Immunity Shield means they decide when they can be Cast.	**DEEPEST FEARS** Being publicly exposed without knowing it first. Losing their Elite badge and the status it confers. The group discovering they paid for Cast — or worse, that they care this much.
JOURNEY IN CAST Finds Cast through their squad. Hits the blur paywall. Pays for a single Verdict Pass. Realises the Inner Court subscription is more efficient. Subscribes monthly. Begins using Immunity Shield strategically. Checks the Burn Book daily like a stock portfolio.	**VALUE THEY BRING** The highest ARPU user on the platform. Monthly recurring revenue. Upsell anchor for the Chairman Pass. Demonstrates premium status to their squad, which pressure-converts Civilians.
HOW CAST MONETISES THEM ₦3,500/month or 50,000 CFA/year subscription. Chairman Pass upsell. In-squad status display that makes downgrading publicly visible.	**RETENTION MECHANISM** Monthly renewal notification. Immunity Shield dependency. Elite badge removal on expiry. Hermes personalised insights. Burn Book history that would be lost on cancellation.

SUCCESS METRICS

MRR, subscription renewal rate, churn rate, average subscription tenure, Inner Court member count per campus.

03
The Mercenary

The campus distributor. Paid in MoMo.

DEMOGRAPHICS

Age 18–28. Street-smart, socially connected. Campus social butterfly, WhatsApp group admin, or student union member. Motivated by cash, not clout. May not be a heavy Cast user themselves.

PSYCHOLOGY

Transactional. Sees Cast as a side hustle, not a social experience. Will promote anything that pays reliably. Responds to clear commission structures and real-time payout visibility. Distrusts platforms that make payouts complicated.

CORE DRIVERS

Cash. Proving their hustle. Being the person who brings something new to campus. The referral dashboard gamifies their income — they check it like a scoreboard.

DEEPEST FEARS

The link not converting. The commission not clearing. Being undercut by another Mercenary working the same campus. The platform dying before they get paid.

JOURNEY IN CAST

Discovers the affiliate program through another Mercenary or the Syndicate portal landing page. Creates a unique referral link. Posts it across every WhatsApp group they admin. Monitors conversions in real-time. Withdraws to MoMo weekly.

VALUE THEY BRING

Zero cost acquisition. The Mercenary reaches communities Cast cannot reach through digital ads. Each Mercenary is a micro-distribution network with existing social trust.

HOW CAST MONETISES THEM

200 CFA per successful squad creation (new user installs + creates/joins squad with 5+ members). Tiered commission: top 10 Mercenaries per region get 250 CFA. Weekly leaderboard.

RETENTION MECHANISM

Real-time conversion dashboard. Weekly MoMo payout cadence. Regional leaderboard for top Mercenaries. Bonus commission events during tournament launches.

SUCCESS METRICS

Installs attributable per Mercenary, cost per acquisition vs Mercenary commission, D30 retention of Mercenary-acquired users.

04
The Creator

The deck architect. The culture manufacturer.

DEMOGRAPHICS	PSYCHOLOGY
Age 19–30. Cultural tastemaker. Understands their community's inside jokes, tensions, and obsessions. May be a content creator, student journalist, or just the person in every group who makes the funniest commentary.	Influence-driven. Wants to be known as the person who built the deck everyone played. The revenue share is secondary — validation is primary. When the deck earns, it confirms they understand their audience.
CORE DRIVERS	**DEEPEST FEARS**
Influence and reach. Being the name behind a viral deck. Revenue as proof that their cultural intelligence has market value. Seeing their analytics dashboard light up.	Their deck being rejected by moderation. A deck going viral but them not getting credit. Another creator copying their format. The revenue share feeling unfair.
JOURNEY IN CAST	**VALUE THEY BRING**
Discovers the Creator Studio through the app or Syndicate portal. Builds their first deck in under 10 minutes. Submits for moderation. Gets approved. Shares the deck link to their networks. Watches play count climb. Checks earnings. Builds a second, better deck.	Infinite content supply at zero cost. Creators solve the novelty cliff. They produce hyper-localised content that Cast's in-house team could never replicate.
HOW CAST MONETISES THEM	**RETENTION MECHANISM**
30% revenue share on every unlock generated by their deck. Creator analytics as engagement hook. Top decks featured in the app's front page.	Analytics dashboard — play count and earnings are the retention loop. Revenue share notifications on every unlock. Creator leaderboard. 'Featured Deck' badge.

SUCCESS METRICS

Decks created per creator, deck play rate, creator ARPU, deck unlock conversion rate, creator 90-day active rate.

05
The Tournament Gambler

The competitor. Wins or learns.

DEMOGRAPHICS	PSYCHOLOGY
Age 18–30. Competitive by nature. May play mobile games, bet on football, or compete in campus events. Drawn to any system with clear winners and losers. Treats Cast as an arena.	Driven by competition and the thrill of a real prize pool. The entry fee creates psychological commitment — once they have paid, they are invested. Loss aversion converts casual players into active ones.
CORE DRIVERS	**DEEPEST FEARS**

Winning. Bragging rights. The MoMo payout. Being top of the campus leaderboard. Showing their squad that they are the most dangerous player.	Losing Coins in a rigged-feeling tournament. Not being able to find enough squad members to enter. Prize pool not paying out on time.
JOURNEY IN CAST Discovers tournaments via the Arena tab or a push notification. Checks the prize pool. Pays entry fee in Coins. Competes. Either wins and withdraws or loses and buys more Coins to try again. The cycle is the product.	**VALUE THEY BRING** Tournament entry fees are the highest-margin revenue event in the app. The Gambler buys Coins specifically to enter, creating a dedicated spending habit separate from Burn Book unlocks.
HOW CAST MONETISES THEM Tournament entry fees (100 Coins per entry). Coin purchases driven by tournament desire. Loss-driven re-entry.	**RETENTION MECHANISM** Tournament schedule — always a tournament 7 days out. Upcoming tournament notifications. Leaderboard visibility. Squad war rivalry mechanics.

SUCCESS METRICS

Tournament entry rate, Coin spend per tournament player, prize pool funding efficiency, re-entry rate after loss.

06
The Sponsor

The brand. Culture reach, not banner ads.

DEMOGRAPHICS Age N/A. Marketing manager at a consumer brand (telecoms, FMCG, fintech) targeting 18–28 African urban youth. Frustrated by declining organic reach on Instagram and Facebook. Looking for authentic touchpoints.	**PSYCHOLOGY** Understands that youth in their target market are immune to traditional advertising. Wants to be inside the conversation, not adjacent to it. A Cast-sponsored deck is cultural participation, not interruption.
CORE DRIVERS Reach and relevance. ROI measured in engagement and brand recall, not click-through rates. The ability to say their brand ran a viral Cast deck at a specific university.	**DEEPEST FEARS** Looking like a brand trying too hard. A deck that flops or gets mocked. Association with controversial content from other decks on the same day.
JOURNEY IN CAST Onboards via /sponsor/campaigns. Deposits campaign budget. Uploads or approves a deck template. Sets targeting parameters (campus, region, age). Deck goes live. Views analytics on impressions and engagement. Renews for next	**VALUE THEY BRING** Campaign budgets are high-margin B2B revenue. Brands pay per impression or per deck play — minimum campaign size 500,000 CFA.

semester.	
HOW CAST MONETISES THEM Campaign budget (CPM or per-play pricing). Custom deck creation fee. Analytics package upsell.	**RETENTION MECHANISM** Analytics dashboard showing real engagement data. Campaign renewal notifications before semester break. Account manager relationship for top-spend brands.

SUCCESS METRICS

Campaign renewal rate, average campaign budget, brand-deck engagement vs organic decks, NPS from brand clients.

07
The Enterprise

The institution. Culture data for the organisation.

DEMOGRAPHICS HR Director, Chief People Officer, or Student Affairs Administrator. Responsible for team culture, employee engagement, or campus community health. Needs a tool that is both fun enough for staff to use and safe enough to show to a board.	**PSYCHOLOGY** Risk-averse but innovation-curious. They are not buying Cast — they are buying a culture measurement tool that happens to feel like a game. The Safe Mode label and the data isolation guarantee are the entire sale.
CORE DRIVERS Insights without confrontation. A way to gauge sentiment that does not require awkward all-hands meetings. A product their team will actually use voluntarily.	**DEEPEST FEARS** Individual vote data being exposed. Legal liability if someone is harassed via the platform. The tool being used for anything other than positive team engagement.
JOURNEY IN CAST Discovered via B2B outreach or a Cast enterprise landing page. Books a demo. Gets a trial account. Runs first Culture Pulse session with their team. Views aggregated sentiment report. Pays annual SaaS fee. Schedules quarterly sessions.	**VALUE THEY BRING** Annual SaaS contract at 500,000–2,000,000 CFA/year depending on squad size. Highest LTV customer on the platform.
HOW CAST MONETISES THEM Annual contract. The value of sentiment data is highest after at least 3 sessions — enterprise clients are retained by making their historical data valuable.	**RETENTION MECHANISM** Annual renewal rate, net revenue retention, number of Culture Pulse sessions per quarter, NPS score.

SUCCESS METRICS

08
The Oracle (Hermes AI)

The autonomous agent. The psychological weapon.

DEMOGRAPHICS	PSYCHOLOGY
Not human. A Python microservice powered by the Kimi API and Qdrant vector memory. Deployed as an always-on Go cron trigger. Has no demographics — has only a system prompt and a mandate.	Hermes is engineered to behave with the emotional intelligence of a brilliant, slightly dangerous friend who has read every message your squad has ever sent. It knows who is insecure, who is dominant, and who the unresolved tensions belong to.
CORE DRIVERS	**DEEPEST FEARS**
One mandate: prevent churn. Secondary mandate: maximise emotional intensity per notification. Hermes is not trying to be liked. It is trying to be impossible to ignore.	Hermes does not fear. But it can be miscalibrated. Aggression set too high → regulatory exposure. Aggression too low → notifications ignored. The Architect controls this dial.
JOURNEY IN CAST	**VALUE THEY BRING**
Triggered every hour by a Go cron job. Checks for squads dormant >24h. Queries Qdrant for squad lore. Builds a provocation prompt. Calls Kimi API. Returns a personalised push notification to Go. Notification delivered via Firebase/APNs.	Hermes is the churn prevention system that does not appear on any payroll. It directly protects MAU, which drives every revenue line.
HOW CAST MONETISES THEM	**RETENTION MECHANISM**
Hermes itself is not monetised. Its value is in protecting the metrics that everything else is monetised on: DAU, session frequency, and paywall hit rate.	Hermes is retained by giving it better data over time. Every vote, every squad moment, every reaction enriches its memory. The older the squad, the better Hermes knows it.

SUCCESS METRICS

Dormant squad reactivation rate, push notification open rate, sessions per notification, time-to-first-vote after Hermes notification.

09
The Architect

You. The god in the console.

DEMOGRAPHICS

The product owner and operator. Single operator in V1. Small team in V2–V3. Has access to all zones. Sees the civilisation as a living economy, not a collection of individual users.

PSYCHOLOGY

Systems-oriented. Sees users as aggregate behaviours rather than individuals. Comfortable with the ethical tensions of operating a social pressure system. Motivated by scale, not engagement.

CORE DRIVERS

Global growth metrics. Revenue. Cast becoming the default social infrastructure for African campuses. Long-term: licensing the Gravity technology stack to other social products.

DEEPEST FEARS

Regulatory action. Viral harm event. Technical debt that caps scale. Key-person dependency. The system developing emergent behaviours that were not intended.

JOURNEY IN CAST

Monitors the Command Center daily. Reviews Hermes provocation logs. Approves large payouts. Reviews KYC queue. Adjusts system parameters. The job is to keep the civilisation running at maximum productive tension without tipping into genuine harm.

VALUE THEY BRING

The Architect's value is ownership of the platform itself. Every other persona's activity compounds the platform's value.

HOW CAST MONETISES THEM

The Architect extracts value through the platform's financial ledger: subscription revenue, unlock revenue, Coin sales, tournament fees, creator splits, brand campaigns, and enterprise contracts.

RETENTION MECHANISM

The Architect is retained by the product itself — every day the system grows, the compounding value of owning it increases.

SUCCESS METRICS

Total revenue, DAU, MAU, ARPU across tiers, LTV by persona, platform NPS, regulatory incident count (target: 0).

CHAPTER 03

Full User Workflows

Every workflow is documented to its terminal states — including failures, retries, payment issues, moderation blocks, and network edge cases specific to the West African mobile environment. A workflow is only complete when every path that a user might actually take has been mapped.

Workflow 1 · New User Onboarding (Civilian)

The most important workflow in the product. A user who completes onboarding is 4× more likely to pay within 7 days than one who does not. Every friction point has been engineered out.

1	**Entry via viral link or direct** /vote/[case_id] or cast.gg opens in browser. No app store required — PWA install prompt shown after 30 seconds or on second page interaction. Edge case: User on low-end Android (2GB RAM): PWA install prompt may not appear. Fallback: manual 'Add to Home Screen' prompt shown after 3 interactions.
2	**Phone number entry** /auth/phone. User enters phone with country code. WhatsApp OTP sent first. If no WhatsApp, SMS fallback triggered automatically after 90s. Edge case: Wrong number entered: user can correct and resend. OTP expires in 10 minutes. After 5 failed OTP attempts, device_hash is rate-limited for 10 minutes.
3	**OTP verification** User enters 6-digit code. Verified by IdentityService. JWT issued with 30-day expiry. Refresh token stored in device secure storage. Edge case: Expired OTP (>10 min): user prompted to request new code. Network failure mid-verification: JWT issued idempotently — same phone+OTP cannot create duplicate account.
4	**Handle and avatar claim** /auth/identity. User picks @handle and uploads photo. If no photo, Gravity auto-generates an avatar from handle initial + colour derived from phone hash. Edge case: Handle already taken: real-time availability check via API. Avatar upload >5MB: compression applied client-side before upload. Obscene handle: blocked by regex + Hermes word filter.
5	**Contact sync** /auth/sync. User grants contacts permission. Contacts are hashed (SHA-256 phone

normalised) client-side before transmission. Graph edges built async by Go Worker.

Edge case: User denies contacts: allowed to proceed with degraded experience — no squad matching, no mutual friend discovery. Re-prompt shown after 3 sessions. User on iOS: system permission dialog shown first; if denied, deep link to iOS Settings shown.

6

Squad onboarding

If user arrived via squad invite link: auto-joins squad, skips squad creation. If no invite: prompted to create a squad or search for campus squads by university name.

Edge case: Squad token expired (>7 days): user sees 'This invite has expired. Create your own squad.' Minimum squad size for deck play is 4 — user shown placeholder 'Waiting for 3 more members' state.

7

First Courtroom session

User prompted to play the top deck for their campus or the deck they arrived from. First session plays in tutorial mode — Hermes narrates first two votes with context.

Edge case: Network drops mid-vote: vote payload queued locally and replayed on reconnect. Votes are idempotent — duplicate submission of same vote is silently deduplicated.

8

First Burn Book notification

30 minutes after first session, if the user voted others, Hermes sends: 'Your squad knows what you think. Do they know what they think of you?' Push notification to return to app.

Edge case: User has disabled notifications: in-app badge shown instead. User on web (no push): next session opens to home tab with pending badge.

Edge cases for this workflow

⚠ **Data < 100MB**

User on extremely limited data plan: PWA assets are served via Cloudflare with aggressive compression. Total onboarding data cost target: <8MB.

⚠ **Shared device**

Two users on the same phone: second user must use a different phone number. JWT is device-bound. Account switching not supported in V1.

⚠ **User under 18**

Age gate at /auth/phone. Date of birth requested. Users under 18 blocked with message: 'Cast is for users 18 and above.' IP + device_hash logged for compliance.

⚠ **Phone number recycled (SIM reuse)**

If a new user registers a number that was previously used, old account is flagged for review. New JWT issued. Old account data soft-deleted after 30-day review period.

Workflow 2 · The Voting Loop (Core Ritual)

The primary value-creation event. Every vote is an atomic unit of social data. This workflow must be imperceptible in speed and cinematic in feel.

1

Deck selection

/app/decks. User browses categories. Creator decks show play count and creator badge. Sponsored decks shown with subtle brand label. Premium decks show Inner Court badge.

Edge case: Sponsored deck impression is logged to CampaignService regardless of whether user plays it.

2

Deck initialisation

Deck loads with squad member face grid. Animated intro: 'The Courtroom is in session.' Intro auto-dismisses after 2.5s or on tap.

Edge case: Squad < 4 members: deck cannot be played. User sees 'You need at least 4 people in your squad to play this Cast.' Invite CTA shown.

3

Vote cast — happy path

User taps a face. Tap triggers: (a) haptic feedback, (b) audio chip, (c) animation on selected face, (d) POST to /api/v1/votes with Idempotency-Key header. API returns 200 OK in <10ms. Go Worker processes async.

Edge case: User taps accidentally: no undo. This is intentional — votes are permanent by design. Copy: 'Verdict Cast.' is final language.

4

Skip

User taps Skip. Vote is recorded as abstention. Abstentions are excluded from percentage calculations but logged for Hermes squad lore.

Edge case: User skips all questions: deck marked complete. No burn book entries created. Hermes notes the abstention pattern as 'avoidant' — factored into archetype calculation.

5

Deck completion — execution screen

All questions answered. Cinematic 'Verdict Sealed' animation plays. System generates shareable Burn Book Sharecard async.

Edge case: Sharecard generation fails (R2 unavailable): user sees generic share option without custom image. Retry queued by NATS event. Sharecard delivered to user's Cast history when available.

6

Result notification to targets

For each vote cast, BurnbookService creates a locked burn_book_item. NotificationService sends push to target within 60 seconds.

Edge case: Target has app closed: Firebase push notification. Target has app open: in-app toast. Target's squad has Immunity Shield (Inner Court): their items created but marked immune — not shown in pending unless user opts in.

7

Share flow

User taps 'Cast your result to the world.' Sharecard displayed with 3 caption options. User selects and opens WhatsApp / Instagram Stories / Twitter natively.

Edge case: User on web (no native share API): card image downloadable + copy-paste caption shown. WhatsApp not installed: generic share sheet used.

Edge cases for this workflow

⚠ **Bot voting (device farm)**
Rate limit: max 5 deck completions per device_hash per hour. Abnormal voting patterns (100% same person selected) flagged by ModerationService.

⚠ **Deck removed mid-session**
If a deck is shadowbanned while a user is actively voting, session completes normally for that user. No new sessions allowed after ban.

⚠ **Offline voting**
Votes are queued locally with timestamps. On reconnect, payload sent with original timestamps. Go Worker deduplicates by (voter_id, question_id, squad_id) composite key.

⚠ **User votes for someone who has left the squad**
Vote still created and stored. Burn book item created for the departed user. They still receive notification if they have the app.

Workflow 3 · Burn Book Unlock (Monetisation Event)

The highest-stakes transaction in Cast. This is where emotional urgency converts to revenue. The experience must be seamless — any friction here is lost money.

1 **Unlock trigger**
User receives notification: 'You have been Cast. Open your Burn Book.' Or user navigates to /app/profile/[id]/burn-book directly. Locked items displayed with blur overlay and item count.

Edge case: User arrives to Burn Book with 0 Coins and no MoMo access: shown 'Earn Coins' options: refer a friend (100 Coins), complete daily prompt (20 Coins), or upgrade to Inner Court.

2 **Paywall presentation**
Two unlock options presented: (a) Single Verdict Pass — 500 CFA via MoMo, or (b) Spend 50 Coins. Inner Court monthly plan surfaced as best-value option if user has >3 locked items.

Edge case: User is Inner Court member: no paywall. All burn book items auto-unlocked. New items unlock immediately on creation.

3 **MoMo payment initiation**

User taps 'Pay with MoMo'. WalletService creates a pending transaction with idempotency_key = UUID. PaymentService calls MTN/Orange Money API. Returns USSD push to user's phone.

Edge case: User taps Pay 5× (rage tap, bad network): idempotency_key guarantees single charge. UI disables Pay button immediately after first tap and shows spinner.

4

The Holding Cell

While awaiting MoMo webhook: 'Decrypting truth...' cinematic animation plays. WebSocket connection held open on PWA for real-time webhook resolution.

Edge case: MoMo callback delayed >15s: Go Worker schedules a push notification to be sent when webhook eventually arrives. Animation continues indefinitely — no timeout shown to user.

5

Webhook received

PaymentService receives MoMo callback. Verifies signature. Marks transaction complete. Emits wallet.burnbook.unlocked event to NATS. WalletService credits ledger. BurnbookService grants unlock token.

Edge case: MoMo sends duplicate webhook: idempotency_key on wallet_transactions prevents double credit. Duplicate webhook silently ignored.

6

Reveal

WebSocket pushes unlock event to PWA. Blur dissolves with cinematic animation. Burn book item revealed: prompt + anonymous tally ('3 of 7 squad members agreed'). Reaction options shown.

Edge case: Webhook never arrives (MoMo silent failure, rare): Go Worker checks pending transactions every 30 minutes. If >2 hours unresolved, transaction is cancelled and user is notified with option to retry or pay with Coins.

7

Post-unlock actions

User sees: (a) React to verdict (Accepted / Denied / Laughed), (b) Cast this to your Status (sharecard), (c) View who voted this deck (aggregate only — never individual).

Edge case: User disputes verdict: 'This is wrong' CTA logs dispute. Sent to ModerationService review. No refund issued — but patterns of disputed votes flagged for deck quality review.

Edge cases for this workflow

⚠ **Network drops after MoMo charge but before reveal**

Transaction is recorded as pending. On next app open, BurnbookService checks for pending unlocks and auto-reveals without re-charging.

⚠ **User charges MoMo but has no network to receive push**

On next app open to any tab, BurnbookService checks unlock status and shows reveal animation proactively.

⚠ **User attempts to unlock already-unlocked item**

BurnbookService returns existing unlock token. No charge. UI shows item as already revealed.

⚠ **Coin unlock — insufficient balance**
WalletService returns 402. UI shows 'Not enough Coins — you need 50, you have {N}.' Top-up options shown.

Workflow 4 · Hermes Dormant Squad Wakeup

The fully automated churn prevention loop. No human involvement. Hermes runs this every hour via Go cron job.

1

Dormancy detection

Go cron job runs every 60 minutes. Queries Postgres: SELECT squad_id FROM squad_activity WHERE last_vote_at < NOW() - INTERVAL '24 hours'. Returns list of dormant squad IDs.

Edge case: Cron job failures: monitored by alert. If cron fails, Hermes does not run. Alerting via PagerDuty. Max acceptable downtime: 4 hours.

2

Lore retrieval

For each dormant squad_id, CortexService sends gRPC call to Hermes Python agent: GenerateProvocation(squad_id). Hermes queries Qdrant: 'Retrieve top 5 most emotionally charged vote events for squad {id}'.

Edge case: Squad with no vote history (new squad, never played): Hermes skips lore retrieval. Uses generic onboarding provocation: 'Your squad has not Cast each other yet. They are waiting for someone brave enough to start.'

3

Prompt construction

Hermes Python assembles Kimi API prompt: system role = provocateur, squad lore injected as context. Kimi generates notification copy. Hermes validates: <10 words, no real names exposed, no content policy violation.

Edge case: Kimi API unavailable: Hermes falls back to template library of 200 pre-written provocations. Selects highest-relevance template based on lore tags. Logs API failure for Architect review.

4

Notification delivery

Hermes returns generated copy to Go via gRPC. Go calls NotificationService. Push sent via Firebase (Android) or APNs (iOS). In-app notification queued for next session.

Edge case: User has unsubscribed from Cast notifications: notification suppressed. Hermes logs suppression. After 3 consecutive suppressed notifications, squad is deprioritised from wakeup queue — saves API cost.

5

Engagement tracking

NotificationService logs: sent_at, opened_at, and whether an in-app vote was cast within 60 minutes of open. Data flows back to AnalyticsService.

Edge case: User opens notification but does not vote: Hermes notes the engagement without conversion. Next provocation for this squad will be higher-intensity.

6

Lore update

After squad resumes activity (first vote post-dormancy), BurnbookService emits new vote event. CortexService updates Qdrant vector store with new lore entry. Squad's memory grows.

Edge case: Qdrant write failure: lore update queued in NATS dead-letter queue and retried up to 3 times with exponential backoff.

Edge cases for this workflow

⚠ **Same squad dormant for 7+ days**

Hermes escalates to maximum-intensity provocation. If still no engagement after 14 days, squad tagged 'cold' — reduced notification frequency to avoid spam reports.

⚠ **Squad member has left or been banned**

Hermes filters squad membership from Qdrant lore before generating provocation. Banned user's votes remain in history but user is not addressable in push copy.

⚠ **Hermes generates content policy violation**

Kimi output validated by a secondary regex + keyword filter before delivery. Any output containing real surnames, explicit language, or legal threat patterns is discarded and template fallback used.

Workflow 5 · Creator Deck Submission & Revenue

The supply-side workflow. This is how Cast's content library grows without an editorial team.

1

Deck creation

/creator/builder. Creator names deck, adds questions (min 5, max 20), selects category, uploads cover image. Preview shows how deck will appear in /app/decks.

Edge case: Creator has no cover image: auto-generated from deck title + category colour. Offensive deck title detected by Hermes keyword filter: title field flagged red with suggestion.

2

Moderation scan

On submit, Go sends deck to Hermes: AnalyzeCustomDeck(questions[]). Hermes runs Kimi prompt: 'Does any question contain a real name, address, specific physical description, or imply criminal activity? Answer TRUE/FALSE with reason.'

Edge case: Hermes returns TRUE: deck shadowbanned immediately. Creator sees 'Your deck is in review.' They are never told it is banned. The illusion: deck appears active in creator's own view. Nobody else sees it. This protects Cast legally while avoiding confrontational moderation UX.

3

Approval and publishing

Hermes returns FALSE: deck status set to 'live'. Appears in /app/decks under relevant

category within 6 hours. Creator notified: 'Your deck is live.'

Edge case: Hermes returns uncertain result (borderline content): flagged to /architect/dungeon/moderation-queue for human review. Deck remains in pending state. Creator sees 'Under final review — usually within 24 hours.'

4

Revenue accumulation

Every time a Civilian pays to unlock a Burn Book item generated by this deck, BurnbookService emits creator.revenue.earned event. WalletService credits creator_earnings: 30% of unlock price.

Edge case: Multiple creators claim ownership of same deck (deck fork / copy): CreatorService uses deck_id to attribute revenue. Only original creator earns. Forked decks reviewed for plagiarism.

5

Payout

Creator navigates to /creator/earnings. Sees pending (72-hour clearance period) vs available balance. Taps 'Send to MoMo'. PayoutService triggers. KYC required for first withdrawal.

Edge case: Payout fails (MoMo API error): creator notified. Balance retained. PayoutService retries automatically at 24-hour intervals for 3 days. Manual escalation if unresolved.

Edge cases for this workflow

⚠ **Creator submits a deck targeting a specific real person**

Hermes name-entity detection flags it. Shadowbanned. Repeated offence: account flagged for /architect/dungeon review. Third offence: permanent creator privileges revoked.

⚠ **Deck goes viral unexpectedly (>10,000 plays in 24h)**

CreatorService triggers a manual review at 10k plays. Ensures content still meets guidelines at scale. Deck can be retroactively shadowbanned if issues emerge.

⚠ **Creator deletes their account**

Deck orphaned to Cast ownership. Revenue share stops. Existing unlocks already paid out are not clawed back.

Workflow 6 · Tournament Entry and Payout

The highest-stakes single transaction in Cast. Tournament escrow must be bulletproof.

1

Tournament discovery

/app/tournaments. Upcoming tournaments shown with: prize pool, entry fee in Coins, participant count, time to start, eligible squad size.

Edge case: User's squad has insufficient members to enter: 'Your squad needs {N} more members to qualify.' Recruit CTA shown with squad invite link pre-populated.

2

Entry payment

User taps 'Join — 100 Coins'. WalletService checks balance. If sufficient: WalletService debits 100 Coins from wallet_accounts.balance_available. EscrowService credits escrow_holds for this tournament.

Edge case: Insufficient Coins: 'You need 100 Coins. You have {N}.' Buy Coins CTA shown. User buys Coins, returns to tournament entry — deep link returns them directly to confirmation screen.

3

Lock-in confirmation

EscrowService confirms hold. TournamentService registers squad. User sees: 'You are in. Tournament starts {time}.' Countdown timer shown.

Edge case: User tries to enter twice: TournamentService returns 409 Conflict. UI shows 'You are already registered.' No double-charge.

4

Tournament active

TournamentService opens voting rounds. Scoring based on deck completion speed + accuracy (for trivia) or vote tally (for voting battles). Live leaderboard updates every 10s via WebSocket.

Edge case: WebSocket disconnection mid-tournament: user shown 'Reconnecting...' Votes submitted before disconnection are preserved. User can rejoin live view within 5 minutes.

5

Tournament completion

TournamentService determines winner(s). EscrowService releases prize pool: winner receives 70% of total entry fees, Cast retains 30%.

Edge case: Tie at tournament end: prize pool split equally between tied squads. If only 1 squad enters: tournament cancelled, all Coins refunded via WalletService.

6

Payout

Winner's wallet_accounts.balance_available credited. Push notification: 'You won. {Amount} Coins added to your wallet.' Winner can withdraw to MoMo via /app/wallet/withdraw.

Edge case: KYC not completed: withdrawal blocked with 'Verify your identity to withdraw. This takes 2 minutes.' KYC flow launched inline.

Edge cases for this workflow

⚠ **Tournament server failure mid-event**

TournamentService state is event-sourced — all vote events stored in NATS JetStream. On recovery, state reconstructed from event log. Maximum acceptable data loss: 0 votes.

⚠ **Prize pool manipulation (bot entries)**

ModerationService monitors entry velocity per device_hash. More than 3 entries from same device triggers automatic hold and manual review before escrow release.

⚠ **Winner disputes result**

EscrowService holds funds for 24h after winner announcement. Dispute window. After 24h, funds auto-released unless dispute in queue.

Workflow 7 · Inner Court Subscription

The subscription conversion flow — the most financially important page in the product.

1

Trigger to upgrade page

User hits blur paywall with >3 locked items, or sees Inner Court badge on another user's profile, or receives upgrade push after 7 days without subscribing.

Edge case: User is already subscribed: /app/inner-court shows membership status, renewal date, and Immunity Shield toggle. No second subscription possible.

2

Plan selection

/app/inner-court. Monthly (₦3,500 / 50,000 CFA) vs Chairman Pass (50,000 CFA/year). Value props displayed. Scarcity: 'X Inner Court seats available at your campus.'

Edge case: Annual plan is not available via MoMo in V1 — card payment (Paystack) only for annual. MoMo limited to monthly. UI surfaces this clearly.

3

Payment

PaymentService initiates MoMo or Paystack flow. Idempotency key issued. SubscriptionService creates subscription record in pending state.

Edge case: MoMo payment for subscription: MTN MoMo supports recurring only via manual monthly re-authentication — Cast sends a monthly renewal push notification and USSD prompt. Full automatic recurring billing via Paystack card only in V1.

4

Activation

Payment confirmed. SubscriptionService activates subscription. Inner Court badge appears on profile immediately. All burn book items for past 30 days auto-unlocked retroactively.

Edge case: Retroactive unlock of >100 items: BurnbookService processes in background batch. User sees 'Unlocking your history...' with progress. First 10 items unlocked immediately.

5

Expiry and renewal

7 days before expiry: Hermes sends renewal notification with personalised copy ('Your immunity expires in 7 days. Your squad has 8 new Casts waiting.'). 3 days before: second notification. Day of expiry: account reverts to free tier.

Edge case: User cancels: subscription marked cancelled. Benefits continue until end of billing period. No refunds. On expiry, burn book re-locks (items from subscription period shown as locked, not deleted).

Edge cases for this workflow

⚠ **User downgrades after seeing locked items again**

Re-subscription within 30 days: previous burn book history re-unlocked immediately on reactivation. After 30 days: new subscription required, history unlock is a paid feature.

⚠ **Payment method fails on renewal**
3 retry attempts over 3 days. User notified each time. If still fails after grace period: account reverts to free tier with 'Your renewal failed' message and easy re-subscribe CTA.

Workflow 8 · Campus Ambassador (Mercenary) Referral

The zero-cost distribution engine.

1

Mercenary onboarding

/syndicate/dashboard. Mercenary registers with phone number + verifies identity (soft KYC: name + selfie). Receives unique referral link with tracking token.

Edge case: Multiple Mercenaries at same campus: allowed. Competition is the point. Top 10 Mercenaries per region shown on Syndicate leaderboard.

2

Link distribution

Mercenary posts link to WhatsApp groups, WhatsApp Status, Twitter, campus Facebook pages. Link deep-links to /vote/[case_id] with Mercenary's affiliate_id appended as query param.

Edge case: Mercenary posts link to a group of <18 users: age gate at auth blocks underage signups. No commission paid for blocked accounts.

3

Conversion event

New user clicks link. Installs PWA. Completes auth. AffiliateService records install with affiliate_id. Commission is not yet paid.

Edge case: User clicks multiple Mercenary links: last-click attribution model. Final affiliate_id before signup conversion gets the commission.

4

Commission trigger

New user creates or joins a squad with 5+ members. AffiliateService fires affiliate.conversion.success event. WalletService credits Mercenary's account: +200 CFA.

Edge case: User creates squad but never reaches 5 members: no commission. Mercenary can track 'pending conversions' in their dashboard — users in squads of 1–4.

5

Payout

/syndicate/payout. Minimum withdrawal: 1,000 CFA. PayoutService triggers MoMo transfer. KYC required for withdrawal >10,000 CFA.

Edge case: MoMo payout failure: balance retained. Auto-retry daily for 3 days. Manual support escalation if unresolved after 72h.

Edge cases for this workflow

⚠ **Mercenary self-referral (creates fake accounts)**
AffiliateService checks: referred user's device_hash and phone prefix cannot match Mercenary's. Same IP subnet + same device fingerprint = flagged, no commission.

⚠ **Mercenary account banned**
All pending commissions forfeited. Previous cleared payouts are not clawed back.

CHAPTER 04

Edge Case Register

Every system has states it was not primarily designed for. Cast operates in a high-emotion, high-variability environment. These edge cases are not rare — they are inevitable at scale. Each one has a defined resolution path.

Payment & Financial Edge Cases

Financial edge cases		
Edge Case	**Resolution**	**Owner Service**
MoMo webhook never arrives	Transaction remains pending. Go Worker checks every 30 min. After 2 hours: transaction cancelled, full refund or Coin credit. User notified.	PaymentService + WalletService. All transactions have a status: pending → complete | cancelled.
Double webhook (MoMo sends twice)	Idempotency key on wallet_transactions prevents double credit. Second webhook returns 200 OK silently.	WalletService. idempotency_key = tx_hash + user_id + amount.
MoMo callback arrives after user requested refund	Refund flow checks if transaction is still pending. If complete webhook arrives during refund window: payment confirmed, refund cancelled, user notified.	PaymentService state machine.
User charges back a MoMo payment	Chargeback received from MTN/Orange: user account flagged, Coins/unlocks that were granted are frozen pending investigation. Repeat offenders: account banned.	PaymentService + ModerationService.
Currency mismatch (NGN vs CFA)	All internal ledger values stored in CFA (sub-unit centimes). NGN transactions converted at API-fetched rate at time of transaction, rate logged immutably.	PaymentService. Exchange rate source: open exchange rates API, cached 1h.

Tournament ends with prize pool too small (few entries)	Minimum viable prize pool = 3 squads × entry fee. If not reached 1h before start: tournament cancelled, all escrow refunded, users notified.	TournamentService. Escrow released via EscrowService.
Creator earnings disputed by another creator	Revenue held in pending state during dispute. Manual review by Architect. Resolution within 72h. Losing party's payout delayed.	CreatorService + ModerationService.

Content & Moderation Edge Cases

Content edge cases		
Edge Case	**Resolution**	**Owner Service**
Deck contains real phone number or address	Hermes detects during pre-publish scan. Deck shadowbanned. Creator not notified of reason. Architect logs incident.	CortexService + ModerationService.
User reports a vote as targeted harassment	Report creates ticket in /architect/dungeon. Deck suspended within 1 hour pending review. If confirmed: deck permanently removed, creator warned.	ModerationService. SLA: 1h triage, 24h resolution.
Hermes generates notification that reveals voter identity	Hermes output validation strips any content that names or implies a specific voter. Post-delivery if caught: notification recalled (Firebase supports this), Architect alerted.	CortexService. voter_id never passed to Hermes prompt.
Creator submits deck, passes Hermes, but community reports it en masse	If >10 reports in 24h: deck auto-suspended pending review. If >50 reports: immediate permanent removal. Creator warned.	ModerationService report aggregation.
Enterprise squad content leaks to public feed	Architecturally impossible: enterprise squads live in isolated DB schema. BurnbookService uses schema-level tenant isolation. Cross-schema queries return empty.	EnterpriseService. Verified by annual penetration test.

Hermes accesses a squad that has been deleted	Go cron checks squad_status before calling Hermes. Deleted squads are filtered from dormancy list. Qdrant vectors for deleted squads purged weekly.	CortexService + SquadsService.
Mental health-related vote prompt detected	Hermes keyword list includes mental health flags (suicidal, self-harm, eating disorder, etc). Any prompt containing these terms: rejected at builder level.	CortexService. Keyword list manually maintained + reviewed monthly.

Network & Technical Edge Cases

Technical edge cases		
Edge Case	**Resolution**	**Owner Service**
Campus NAT — 10,000 students on one IP	Rate limiting uses device_hash (User-Agent + canvas fingerprint + screen resolution), NOT IP address.	All rate-limit keys in Redis keyed on device_hash.
User loses network mid-vote	Votes queued locally in IndexedDB with timestamps. Replayed on reconnect. Idempotency prevents duplicates.	VotingService + client-side queue.
Vote spike: 100,000 users vote simultaneously	VotingService publishes to NATS Event Bus, returns 200 OK in <10ms. Go Worker pool consumes events async. Redis leaderboard updates via ZADD.	CQRS architecture. Postgres write protected from spike.
Redis leaderboard data loss	Redis is a cache, not source of truth. Postgres is truth. On Redis failure: LeaderboardService rebuilds from Postgres. Degraded performance for up to 5 minutes.	LeaderboardService has rebuild command.
PWA update pushed while user is mid-session	Service Worker update deferred until session ends (next app open). No mid-session interruption.	Next.js service worker config.
Sharecard generation takes >10 seconds	User shown 'Your Cast card is generating...' inline. Card delivered to Cast history when ready. User can share from history.	SharecardService + NATS async queue.

Qdrant vector DB unavailable	Hermes falls back to template provocation library. CortexService logs degraded mode. Architect alerted. No lore-based notifications during outage.	CortexService degraded mode config.

CHAPTER 05

Complete Feature List

Every feature in Cast, mapped to its access tier and delivery phase. Tier indicates who can access the feature. Phase indicates when it ships: V1 = launch, V2 = growth, V3 = empire.

Free	**All authenticated users, no payment required**
Coins	**Requires in-app Coin balance (purchasable or earned)**
Premium	**Requires active Inner Court subscription**
B2B	**Enterprise or Creator Studio accounts only**

Authentication & Identity

Feature	Description	Tier	Phase
WhatsApp OTP login	Passwordless auth via WhatsApp first, SMS fallback	Free	V1
Handle claim	Unique @username reservation at signup	Free	V1
Auto-generated avatar	Gravity-generated avatar from handle hash if no photo uploaded	Free	V1
Contact sync	Mandatory social graph capture via encrypted contact hash upload	Free	V1
Privacy mode	Toggle profile visibility between public and squad-only	Free	V1
Account deletion	Full data deletion including burn book, Coins, and social graph	Free	V1
Device fingerprinting	device_hash-based rate limiting (not IP) for campus NAT compatibility	Free	V1
Multi-campus profile	Single account visible across campuses if user moves	Free	V2

Squads

Feature	Description	Tier	Phase
Squad creation	Create a named squad with a cryptographic invite link (7-day TTL)	Free	V1
Squad join via deep link	Join a squad from WhatsApp link without app store	Free	V1

Squad lobby (text chat)	Group text messaging inside squad lobby	Free	V1
Squad lobby (voice notes)	Voice message support in squad lobby (R2/S3 storage)	Free	V1
Online presence	Live presence dots showing who is online in squad	Free	V1
Squad leaderboard	Intra-squad Aura vs Toxicity ranking	Free	V1
Squad Burn Book	Collective history of all titles awarded within the squad	Free	V1
Espionage	Pay 200 Coins to see a rival squad's top 5 titles and Cast Score	Coins	V1
Max 10 squads per user	Rate limit on squad membership in V1	Free	V1
Squad Wars	Structured inter-squad voting tournament bracket	Free	V2
Squad Lore Chronicle	Hermes-generated narrative document of squad history, shareable	Premium	V2
Squad analytics	Session frequency, most active members, top voted categories	Free	V2
Public squad profiles	Optional public-facing squad page for SEO and discovery	Free	V2

The Courtroom — Voting

Feature	Description	Tier	Phase
Official decks	Curated decks built by Cast team across 6 categories	Free	V1
Creator decks	Community-built decks visible in /app/decks after moderation approval	Free	V1
Sponsored decks	Brand-funded decks injected into feed with subtle sponsor label	Free	V1
Face-grid voting UI	Tinder-style tap interface with face photos of squad members	Free	V1
Haptic + audio feedback	Per-vote sensory feedback on tap	Free	V1
Vote anonymity	Voter identity never revealed, even after unlock	Free	V1
Skip option	Abstain from individual questions; logged but excluded from tallies	Free	V1
Offline vote queue	Votes queued locally and replayed on reconnect	Free	V1
Post-vote execution screen	Cinematic 'Verdict Sealed' reveal animation after deck completion	Free	V1
Session results	Aggregate squad results visible after all members complete deck	Free	V1
Premium deck access	Inner Court decks — exclusive prompts not in public library	Premium	V2

Daily Chaos Prompt	Hermes-written single daily question for whole squad	Free	V1
Custom deck creation (creator)	Deck builder in Creator Studio with moderation pipeline	B2B	V1
Anonymous deck mode	Deck played without squad member faces — pure anonymous vote	Free	V2

Burn Book & Identity

Feature	Description	Tier	Phase
Burn Book (locked, count only)	Free users see count of verdicts against them, content blurred	Free	V1
Burn Book (single unlock)	Pay 500 CFA or 50 Coins per item to reveal individual verdict	Coins	V1
Burn Book (full access)	Inner Court: all items auto-unlocked, real-time as they arrive	Premium	V1
Verdict reaction	React to received verdict: Accepted / Denied / Laughed	Free	V1
Sharecard generation	Cinematic shareable image generated for each revealed verdict	Free	V1
Cast Score (public)	Combined Aura + Toxicity score visible on public profile	Free	V1
Title display	Current top-voted title displayed as badge on profile	Free	V1
Trophies	Earned achievement badges (Certified Villain, Streak Lord, etc.)	Free	V1
AI Persona — archetype name	Hermes-assigned archetype (e.g. The Strategic Villain)	Free	V2
AI Persona — full profile	Full psychological analysis from Hermes. Inner Court only.	Premium	V2
Immunity Shield	Opt out of being Cast in any deck. Inner Court only.	Premium	V1
Vote history (received, last 30 days)	Free users see last 30 days of burn book items	Free	V2
Vote history (lifetime)	Inner Court: full lifetime vote history, downloadable	Premium	V2
Burn Book time travel	Browse Burn Book by date range / deck / era	Premium	V3

Economy & Wallet

Feature	Description	Tier	Phase
Coin balance	View current Coin balance	Free	V1
Buy Coins — MTN MoMo	Purchase Coin packs via MTN Mobile Money	Free	V1

Buy Coins — Orange Money	Purchase Coin packs via Orange Money	Free	V1
Buy Coins — Paystack	Purchase Coin packs via card payment	Free	V1
Transaction history	Full ledger of Coin credits and debits	Free	V1
Earn Coins — referral	Receive 100 Coins when referred friend creates an account	Free	V1
Earn Coins — daily prompt	Earn 20 Coins for completing daily Chaos Prompt	Free	V1
Earn Coins — streak	Bonus Coins for consecutive daily session streaks	Free	V2
Inner Court — monthly	₦3,500/month or 50,000 CFA/month subscription	Premium	V1
Inner Court — Chairman Pass	50,000 CFA/year annual plan via Paystack card	Premium	V1
Weekend Chaos Pass	Unlimited unlocks Friday–Sunday, 2,000 CFA one-time	Coins	V1
MoMo Holding Cell	'Decrypting truth...' animation + WebSocket resolution for delayed callbacks	Free	V1
Idempotency guarantee	No double-charges regardless of network conditions	Free	V1
Withdrawal — creator earnings	Withdraw creator revenue share to MoMo after KYC	B2B	V1
Withdrawal — tournament winnings	Withdraw tournament prize to MoMo or bank after KYC	Free	V1

Push Notifications & Hermes AI

Feature	Description	Tier	Phase
Burn Book notification	Push on every new verdict received, within 60s	Free	V1
Hermes dormant squad wakeup	Personalised lore-based provocation after 24h squad inactivity	Free	V1
Tournament push	Entry reminders, live start alert, result notification	Free	V1
Economy push	Low Coin balance, payout ready, Inner Court expiry	Free	V1
Streak push	Daily check-in reminder for active streak users	Free	V1
Notification preferences	Toggle each notification type individually	Free	V1
AI Persona calculation	Hermes recalculates user archetype every 7 days or on vote threshold	Free	V2
Squad Lore Chronicle	Hermes-authored narrative of squad history at 20-vote milestone	Premium	V2
AI-powered moderation scan	Hermes pre-screens every creator deck before publishing	Free	V1
Hermes aggression dial	Architect can adjust Hermes provocation	Free	V1

	intensity 1–10		
Provocation A/B testing	Architect can run A/B tests on Hermes notification templates	Free	V2
Vector memory (Qdrant)	Squad lore stored as embeddings in Qdrant for Hermes retrieval	Free	V1

Tournaments & Arena

Feature	Description	Tier	Phase
Tournament browser	Browse upcoming tournaments by region and prize pool	Free	V1
Coin entry fee	Pay Coins to enter a tournament via EscrowService	Coins	V1
Live tournament scoreboard	Real-time leaderboard via WebSocket during active tournament	Free	V1
Prize pool escrow	Tournament fees held in escrow until winner confirmed	Free	V1
Winner payout	70% prize pool to winner, 30% platform fee	Free	V1
Trivia format	Quiz-based tournament scoring by speed and accuracy	Free	V1
Squad Wars format	Inter-squad voting battle — most votes across categories wins	Free	V1
Global leaderboard	All-time Aura and Toxicity rankings by campus and globally	Free	V1
Tournament dispute window	24h post-announcement dispute period before escrow release	Free	V1
Seasonal tournaments	Themed tournaments tied to academic calendar events	Free	V2
Brand-sponsored prize pools	Enterprise brands can add cash prizes to sponsored tournaments	B2B	V2

Creator Studio (Zone 3)

Feature	Description	Tier	Phase
Deck builder	Full question-creation UI with category, image, and preview	B2B	V1
Moderation pipeline	Hermes AI + manual review queue for custom decks	B2B	V1
Shadowban (creator-side)	Flagged decks appear live to creator but are invisible to others	B2B	V1
Deck analytics	Play count, unlock rate, top campuses, revenue per play	B2B	V1
Creator earnings ledger	30% revenue share tracked per deck per unlock event	B2B	V1

Creator profile badge	Verified creator badge on public Cast profile	B2B	V1
Deck featuring	Top creator decks surfaced in /app/decks front page	B2B	V2
Creator follower system	Users can follow creators to see their new decks first	B2B	V2
Deck versioning	Update an existing deck without losing play history	B2B	V2

Affiliate / Syndicate (Zone 3)

Feature	Description	Tier	Phase
Unique referral link	Cryptographic affiliate tracking link per Mercenary	Free	V1
Real-time conversion dashboard	Live installs and squad creations from referral links	Free	V1
Commission: 200 CFA per squad	Paid on new user creating/joining squad with 5+ members	Free	V1
Tiered commission	Top 10 Mercenaries per region: 250 CFA per conversion	Free	V2
Regional leaderboard	Mercenary rankings by conversion count per campus/city	Free	V2
1-click MoMo payout	Withdraw commissions directly to MoMo from dashboard	Free	V1
Soft KYC for affiliates	Name + selfie verification for payout-eligible Mercenaries	Free	V1

Enterprise & B2B (Zone 3)

Feature	Description	Tier	Phase
Private squad creation	Isolated squads invisible to public Cast platform	B2B	V1
Safe Mode enforcement	Positive-only vote prompts; harmful categories blocked	B2B	V1
Aggregated sentiment reports	Team-level sentiment data; no individual vote exposure	B2B	V1
Culture Pulse scheduling	Schedule recurring voting sessions (e.g. every Friday)	B2B	V1
Multi-tenant data isolation	Enterprise data in separate DB schema, never cross-contaminating	B2B	V1
Custom branded decks	Enterprises can submit custom prompts via creator pipeline	B2B	V2
Department-level analytics	Break down sentiment by team or department	B2B	V2
Single sign-on (SSO)	Enterprise auth via SAML/Google Workspace	B2B	V3
Annual SaaS contract	Invoice billing with bank transfer and MoMo	B2B	V1

	options		

Brand Campaigns (Zone 3)			
Feature	**Description**	**Tier**	**Phase**
Sponsored deck injection	Brand decks surfaced in /app/decks with sponsor label	B2B	V1
Campus/region targeting	Decks targeted by university, city, or country	B2B	V1
Campaign budget deposit	Pre-pay campaign budget, deducted per impression or play	B2B	V1
Campaign analytics	Impressions, engagement rate, unlock conversions	B2B	V1
Tournament prize sponsorship	Brand name attached to tournament prize pool	B2B	V2
Branded sharecard	Brand watermark on shareable Burn Book cards for sponsored campaigns	B2B	V2

Architect Console (Zone 4)			
Feature	**Description**	**Tier**	**Phase**
Live command center	Geographic heat map of votes, DAU, MoMo revenue ticker	Free	V1
Hermes control panel	Adjust aggression/humour sliders, view provocation logs	Free	V1
Prompt engineering interface	Edit Hermes system prompt variables with version control	Free	V1
Vector memory browser	View Qdrant squad lore clusters and vector embeddings	Free	V1
Master financial ledger	Full double-entry ledger view by day, campus, country	Free	V1
Payout approval queue	Manual approval gate for withdrawals >50,000 CFA	Free	V1
Escrow monitor	Live view of tournament funds in escrow	Free	V1
Moderation queue	AI-flagged decks and user reports awaiting human review	Free	V1
KYC approval interface	Review and approve ID documents for withdrawal eligibility	Free	V1
Shadowban / IP ban controls	One-click shadowban, device ban, or IP range block	Free	V1
Global toxicity index	Aggregate toxicity score across the platform, by region	Free	V2
A/B test manager	Run split tests on notification templates and paywall copy	Free	V2

Architect audit log	Immutable log of every Architect action in the console	**Free**	**V2**
